

Unnamed_Unit

3.Fundamental Theorem of arithmetic units

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Fundamental Theorem of Arithmetic

The fundamental theorem of arithmetic says that "factorization of every composite number can be expressed as a product of primes irrespective of the order in which the prime factors of that respective number occurs". The fundamental theorem of arithmetic is a very useful method to understand the prime factorization of any number.

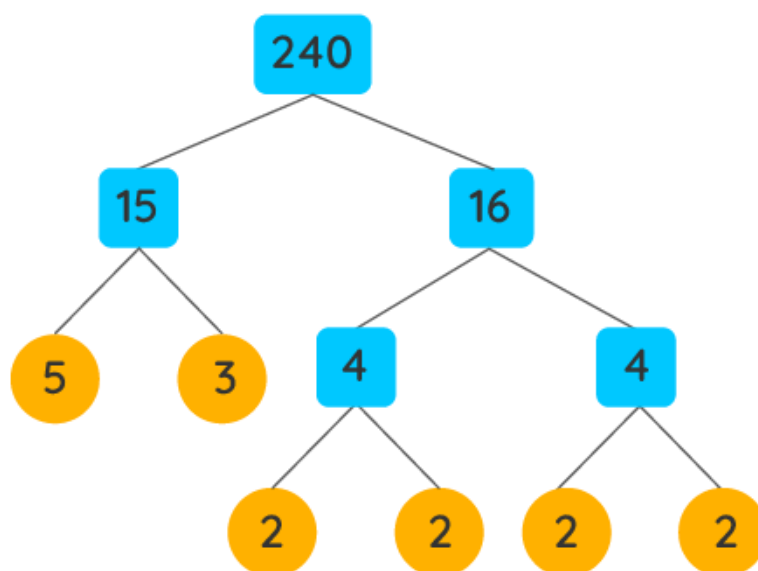
Fundamental Theorem of Arithmetic Definition

The statement of the fundamental theorem of [arithmetic](#) is: "Every [composite number](#) can be factorized as a product of primes, and this factorization is unique, apart from the order in which the prime factors occur."

For example, let us find the prime factorization of 240.

Fundamental Theorem Of Arithmetic

Prime factorization of 240



From the above figure, we get $240 = 2 \times 2 \times 2 \times 2 \times 3 \times 5$. This theorem further tells us that this factorization must be unique. That is, there is no other way to express 240 as a product of primes. Of course, we can change the order in which the prime factors occur. For example, the prime factorization can be written as: $240 = 3^1 \times 2^4 \times 5^1$ or $3^1 \times 2^2 \times 5^1 \times 2^2$ etc. But the set of prime factors (and the number of times each factor occurs) is unique. That is, 240 can have only one possible prime factorization, with four factors of 2 that is 2^4 , one factor of 3 that is 3^1 , and one factor of 5 that is 5^1 .

Fundamental Theorem of Arithmetic Proof

To prove the fundamental theorem of arithmetic, we have to prove the existence and the uniqueness of the prime factorization. Thus, the fundamental theorem of arithmetic proof is done in two steps. We will prove that for every [integer](#), $n \geq 2$, it can be expressed as the product of primes in a unique way: $n = p_1 \times p_2 \times \dots \times p_i$.

Step 1 - Existence of Prime Factorization

We will prove this using [mathematical induction](#).

Basic Step: The statement is true for $n = 2$.

Assumption Step: Let us assume that the statement is true for $n = k$.

Then, k can be written as the product of primes.

Induction Step: Let us prove that the statement is true for $n = k + 1$.

If $k + 1$ is prime, then the case is obvious.

If $k + 1$ is NOT prime, then it definitely has some prime factor, say p .

Then $k + 1 = pj$, where $j < k \rightarrow (1)$

Since $j < k$, by the "inductive step", k can be written as the product of primes.

Thus, from (1), $k + 1$ can also be written as the product of primes. Thus, by the mathematical induction, the "existence of factorization" is proved.

Step 2 - Uniqueness of Prime Factorization

Let us assume that n can be written as the product of primes in two different ways, say,

$$n = p_1 p_2 \cdots p_i, \text{ or,}$$

$$n = q_1 q_2 \cdots q_j$$

Since these are prime factorization, $q_1, q_2, \dots, q_j$ are [coprime numbers](#) (as they are [prime numbers](#)).

Therefore, by [Euclid's lemma](#), p_1 divides only one of the primes.

Note that q_1 is the smallest prime and so $p_1 = q_1$.

In the same way, we can prove that $p_n = q_n$, for all n .

Hence, $i = j$.

Thus, the prime factorization of n is unique.

HCF and LCM Using Fundamental Theorem of Arithmetic

To find the HCF and LCM of two numbers, we use the fundamental theorem of arithmetic. For this, we first find the prime factorization of both numbers. Next, we consider the following:

- HCF is the product of the smallest power of each common prime factor.
- LCM is the product of the greatest power of each common prime factor.

For example, let's find the HCF of 850 and 680. For this, first, we will find the prime factorization of these numbers.

$$\text{Prime factorization of } 850 = 2^1 \times 5^2 \times 17^1$$

$$\text{Prime factorization of } 680 = 2^3 \times 5^1 \times 17^1$$

HCF is the product of the smallest power of each common prime factor. Hence, $\text{HCF}(850, 680) = 2^1 \times 5^1 \times 17^1 = 170$.

LCM is the product of the greatest power of each common prime factor. Hence, $\text{LCM}(850, 680) = 2^3 \times 5^2 \times 17^1 = 3400$.

Thus,

$$\text{HCF}(850, 680) = 170$$

$$\text{LCM}(850, 680) = 3400$$